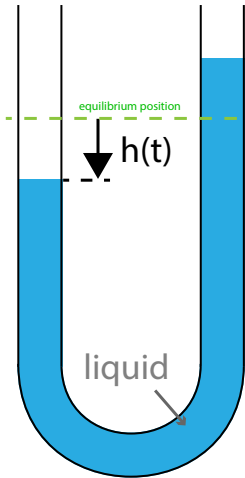


a manometer, used to measure pressure, is a U-shaped tube with liquid trapped inside. both ends of the tube are open.

suppose the right-hand side of the tube is always exposed to the atmosphere. as for the left side, it was exposed to a pressurized gas line for $t < 0$. but, at $t = 0$, the line to the high-pressure gas was disconnected. thus, the left side is exposed to the atmosphere as well for $t \geq 0$. consequently, the liquid in the tube at $t = 0$ initializes (a) at rest and (b) with a taller height on the right side.

our goal is to model $h = h(t)$ [m], the liquid level with respect to the equilibrium position. by convention, $h > 0$ implies the liquid level is lower on the left side.



1. derive a dynamic model for $h = h(t)$ [m] for $t \geq 0$ by writing a force balance (mass $\times$ acceleration = sum of forces) on the liquid. account for (i) the gravitational force and (ii) the viscous forces internal to the liquid. the [constant] parameters are listed in the table below. consider

- the mass of liquid in the tube, which has a circular cross-section, is $\rho\pi R^2 L$ [kg].
- the restoring force from gravity, due to displaced liquid, is $\pi R^2 \rho g 2h(t)$ [N].
- the viscous force, which slows acceleration is $8\pi\mu L \frac{dh}{dt}$ [N] (the Hagen-Poiseuille equation).

you should arrive at a second-order ordinary differential equation (ODE) for $h(t)$.

parameter	description
manometer setup	
R	radius of the tube [m]
L	total length of liquid in tube [m]
properties of the liquid	
ρ	density [kg/m ³]
μ	viscosity [N·s/m ²]
a physical constant	
g	standard acceleration due to gravity [m/s ²]

2. write the second-order ODE for $h(t)$ as a system of coupled, first-order ODEs in matrix-vector form:

$$\frac{d\mathbf{x}(t)}{dt} = A\mathbf{x}(t).$$

the coefficient matrix A should be written in terms of the parameters R , μ , ρ , etc.

3. suppose the manometer tube (on earth, $g=9.8\text{m/s}^2$) has $R = 0.01\text{m}$ and $L = 1\text{m}$; olive oil ($\rho = 997\text{kg/m}^3$, $\mu = 0.001\text{Pa}\cdot\text{s}$) inside; and begins at rest with $h(0) = 0.2\text{m}$. compute the eigenvalues and eigenvectors of the coefficient matrix A in Julia (they should be complex). based on these eigenvalues and eigenvectors, write the solution to $\mathbf{x}(t)$ in the time domain. plot $h(t)$ in Julia for $t \in [0, 60]\text{s}$. for writing out $h(t)$, you may use only one significant figure from the eigenvalues/vectors.